

Implementation Plan

Event Guest Management System

version 1.0



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Contents

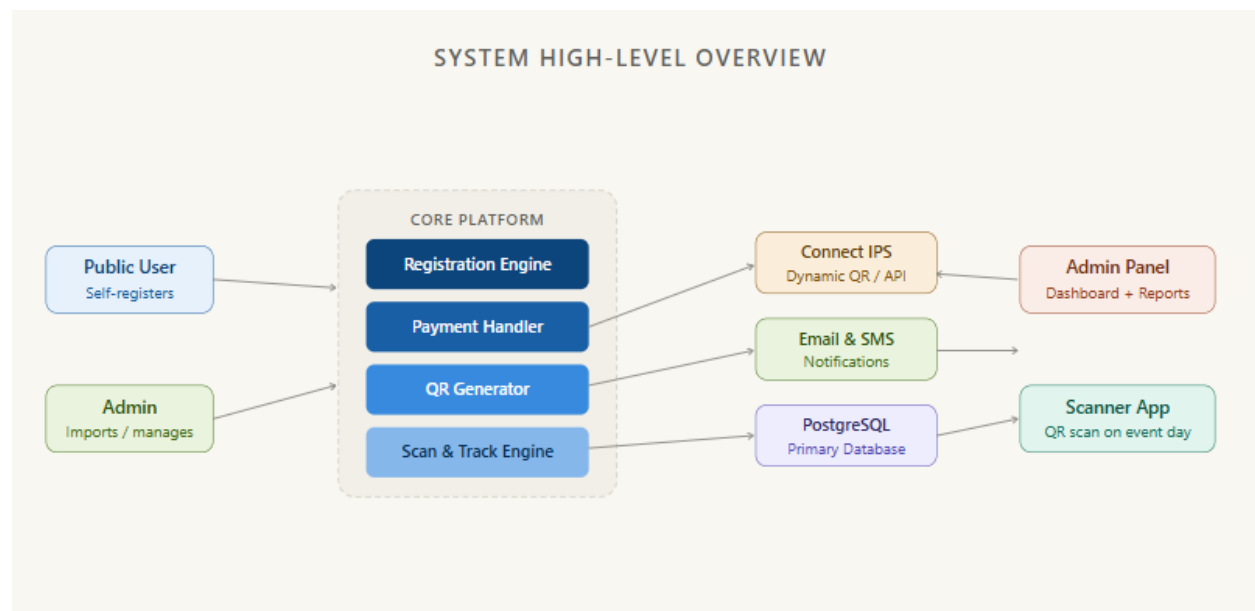
1. System Goal	1
2. Core Modules.....	2
2.1. Event Creation & Configuration Module.....	2
2.2. Participant Category Management.....	3
3. Registration Flows	4
3.1. Self-Registration Flow.....	4
3.2. Admin Imported Registration Flow	6
4. Payment Integration Plan	7
4.1. Connect IPS Dynamic QR / API.....	7
5. Unique QR Code and Guest Number Logic	9
6. Tracking & Scan Module	11
7. Email, Ticket, and SMS Notifications	12
8. Label / Sticker Printing Module	14
9. Dashboard & Admin Panel	15
10. Reports & Exports	17
11. Suggested Database Plan	18
12. Recommended Architecture	27
13. Admin Workflow Example	28
14. Public User Workflow Example.....	28
15. Security Plan	29

1. System Goal

Build a web-based **multi-event event management platform** where admins can:

- create new events
- define participant categories and ticket prices
- enable self-registration with payment
- import non-paying participants via CSV
- generate unique QR-based passes for each event participant
- track actions like check-in, lunch, dinner, card delivery, etc.
- if event is multiple day event track for multiple days
- print labels/badges
- send confirmation by email and SMS
- monitor everything from one admin dashboard
- export reports anytime

This should be designed as a **reusable platform**, not just for one event.



2. Core Modules

2.1. Event Creation & Configuration Module

Admin creates each new event from the admin panel.

Event fields

- Event Name
- Event Code / Slug
- Event Description
- Venue
- Start Date / End Date
- Registration Open / Close Date
- Event Banner / Logo
- Contact Info / Main Organizer
- Status: Draft / Published / Closed / Archived

Admin-configurable settings per event

- enable/disable self-registration
- enable/disable payment
- enable/disable CSV import
- enable/disable check-in tracking
- enable/disable lunch tracking
- enable/disable dinner tracking
- enable/disable card delivery tracking
- enable/disable label printing
- enable/disable email/SMS notifications

This makes the system flexible for future events.

2.2. Participant Category Management

Categories must be fully dynamic from admin panel.

Default sample categories

- General Attendee
- VIP
- Chief Guest
- Organizer
- Volunteer
- Sponsor
- Media
- Speaker
- Exhibitor

Each category should support

- category name
- description
- paid/free
- ticket price
- currency
- label color / badge style
- priority / sort order
- QR access permissions if needed
- active/inactive

Admin should be able to:

- add new category
- edit category
- delete category

- change price later
- assign custom workflows per category

For example:

- VIP may get lunch+dinner+special lounge
- Volunteer may not need payment
- Sponsor may be imported by admin only

3. Registration Flows

3.1. Self-Registration Flow

Used for public users who register themselves.

Steps

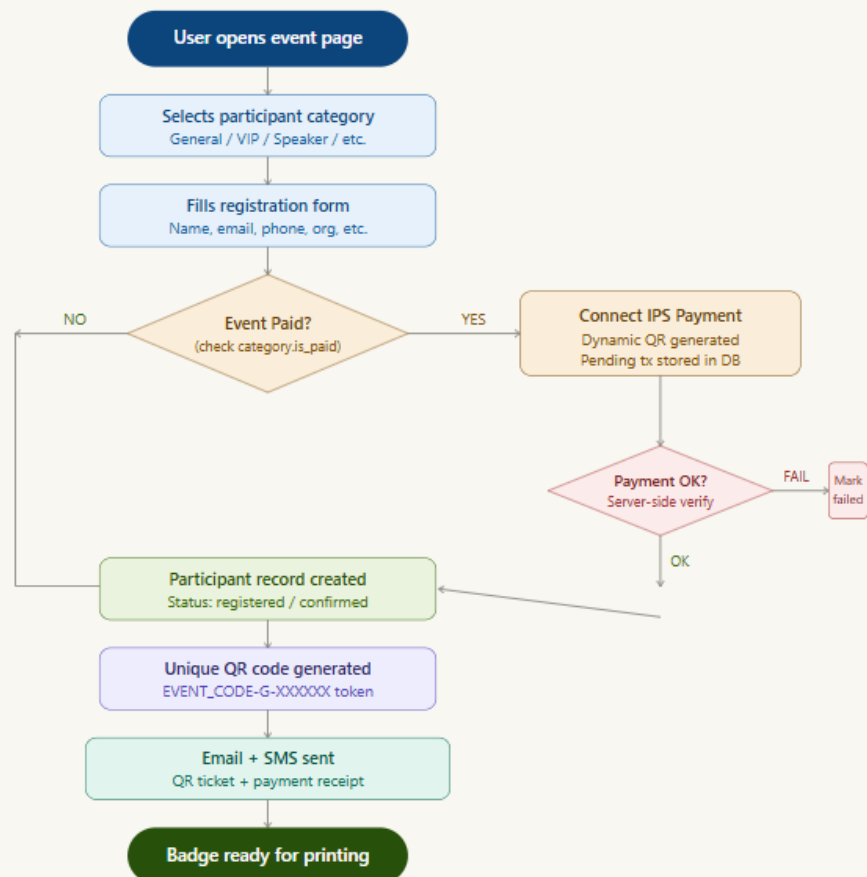
1. User opens event registration page
2. Selects participant category
3. Fills registration form
4. Check if the event is free or paid
5. System calculates ticket fee based on category
6. Payment initiated through **Connect IPS dynamic QR/API**
7. Payment success/failure callback handled
8. Participant record created/updated
9. Unique QR ticket generated
10. Confirmation email sent
11. SMS/phone notification sent
12. Badge/label ready for printing from admin panel

Registration form sample fields

- Full Name
- Email

- Phone Number
- Designation
- Organization
- Address / City
- Category
- Gender (optional)
- Notes / Remarks
- PAN/VAT (if invoice needed)
- Consent checkbox
- Terms acceptance

SELF-REGISTRATION & PAYMENT FLOW



Optional:

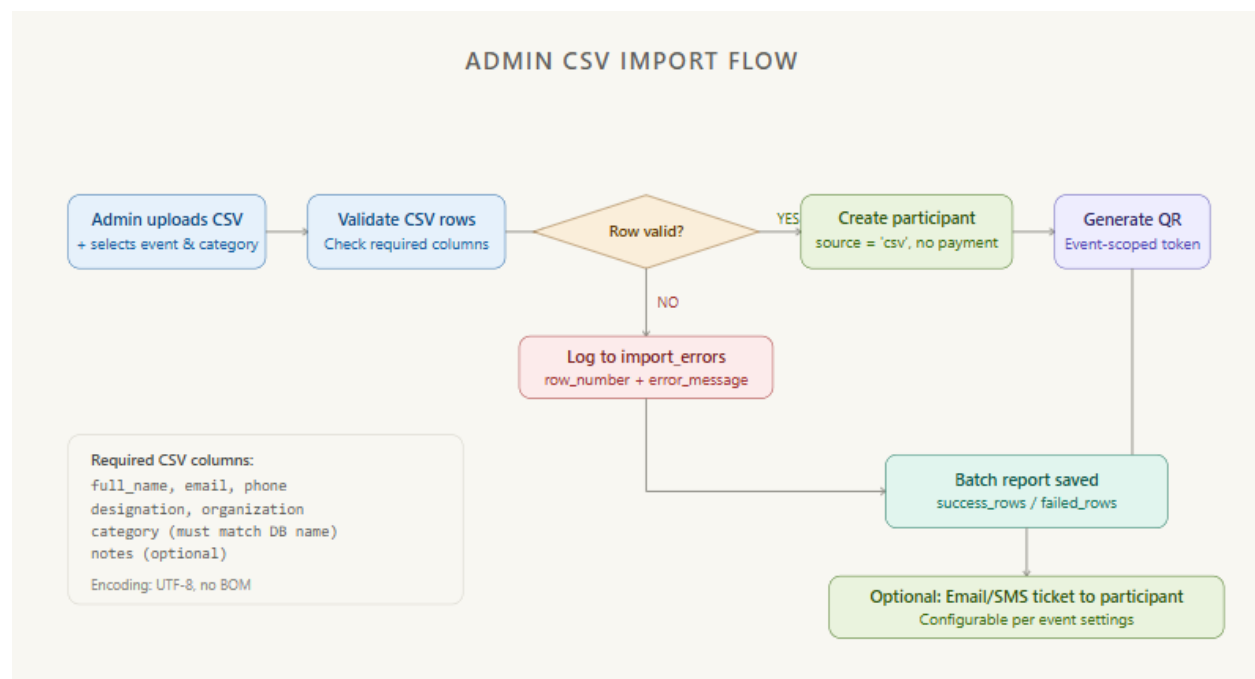
- photo upload
- company name
- meal preference
- special assistance requirement

3.2. Admin Imported Registration Flow

Used for non-self-registration participants.

Use cases

- VIP list
- Chief Guests
- Sponsors
- Organizers
- Volunteers
- Invited guests
- Complimentary participants



Method

Admin uploads CSV file with selected event and category.

CSV sample columns

- full_name
- email
- phone
- designation
- organization
- category
- notes

System behavior

- no payment required
- participant created directly
- unique event-specific QR generated
- ticket/pass can still be emailed/SMSed if needed
- visible in admin dashboard same as normal attendees

Very important: both self-registered and imported participants should ultimately live in the **same participant table structure**, only with different registration/payment source.

4. Payment Integration Plan

4.1. Connect IPS Dynamic QR / API

Since prices differ by category and event, payment must be generated dynamically.

Payment logic

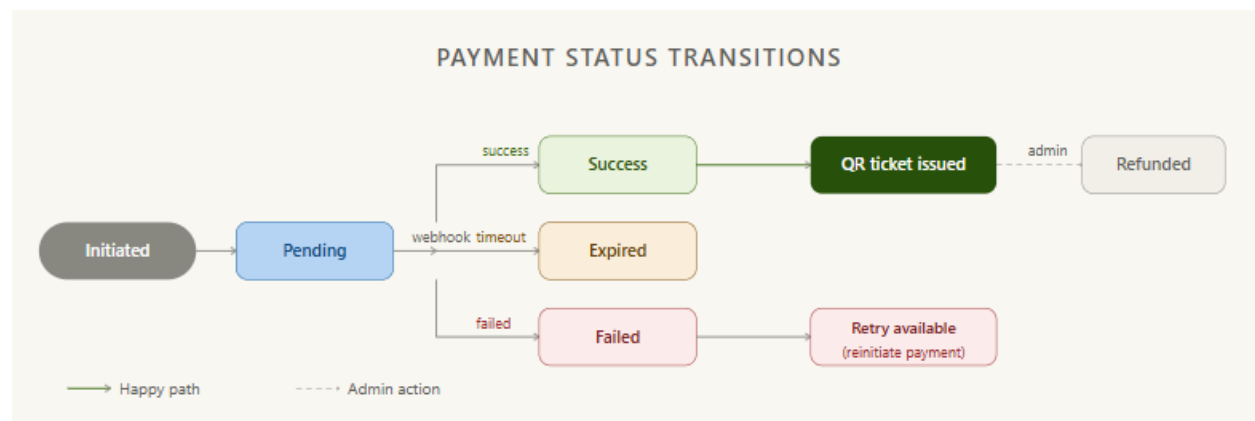
For each self-registration:

- fetch event
- fetch selected category

- get price from admin-defined category pricing
- create payment order
- request Connect IPS dynamic QR / transaction session
- store pending transaction
- user completes payment
- callback/webhook verifies payment status
- mark payment as successful/failed
- issue ticket only after successful verification

Payment statuses

- pending
- initiated
- success
- failed
- cancelled
- expired
- refunded



Important implementation points

- never trust frontend payment confirmation alone
- verify via backend callback/webhook/API status check

- store transaction reference from Connect IPS
- support retry payment for pending/failed transactions
- support admin-side manual verification if needed

Optional Features for first phase

- early bird pricing
- promo code / coupon
- group registration pricing
- tax/VAT handling
- invoice/receipt generation
- payment expiry timer

5. Unique QR Code and Guest Number Logic

Each participant must have a **unique QR code for one specific event only**.

Recommended approach

Generate:

- a unique internal participant ID
- a public-facing random guest code/ticket number

Example:

- Event Code: DNC2026
- Guest Number: G-8F4K29
- Final Ticket Code: DNC2026-G-8F4K29

QR should encode

Not just name directly. Better to encode a secure token or ticket URL such as:

- participant token
- event ID
- unique guest code

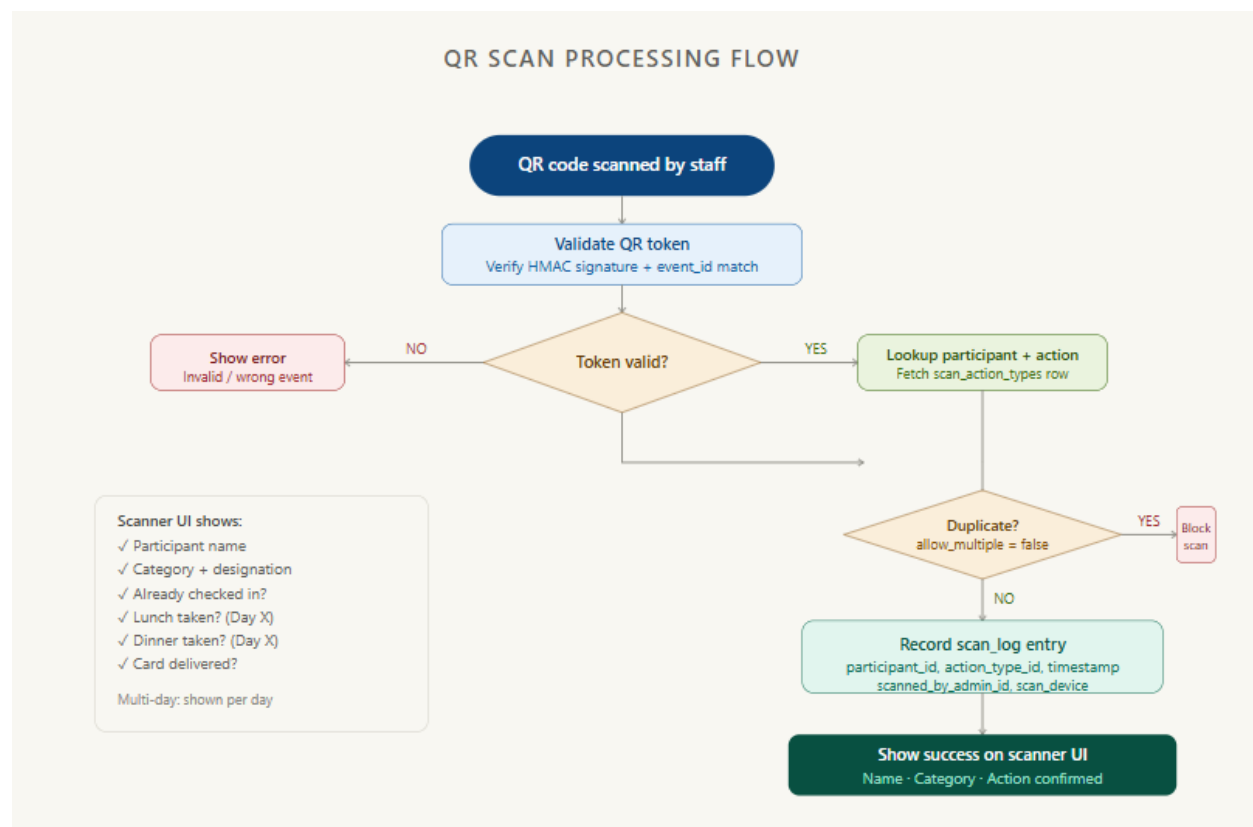
Example QR destination:

https://systemdomain.com/checkin/t/{secure_token}

This is better than storing plain text because it is more secure and easier to validate.

QR should be usable for

- check-in
- lunch taken
- dinner taken
- card delivered
- badge printed
- attendance logs
- custom actions later



6. Tracking & Scan Module

This is one of the most important parts.

Admin/custom tracking actions

System should allow per event configurable scan actions.

Default actions

- Check In
- Card Delivered
- Lunch Taken
- Dinner Taken

Future custom actions

- Session Entry
- Kit Collection
- Parking Pass Issued
- Certificate Collected

Instead of hardcoding only lunch/dinner, design a **custom scan action system**.

Scan record behavior

When QR is scanned:

- identify participant
- identify event
- identify action type
- record scanner/admin user
- record timestamp
- record date/time
- prevent duplicate scans if configured

Example duplicate rules

- Check-In: allow only once
- Lunch Taken: allow once
- Dinner Taken: allow once
- Card Delivered: allow once
- Session Entry: can allow multiple times if needed

Scan interface

Scanner panel should show:

- participant name
- category
- designation
- event name
- already checked-in? yes/no
- lunch already taken? yes/no
- dinner already taken? yes/no
- consider for multiple days event also
- remarks

7. Email, Ticket, and SMS Notifications

Automatic Email after successful registration/payment

Send:

- participant name
- event details
- ticket number
- QR code
- payment confirmation

- invoice/receipt if needed
- venue/time instructions

Email types

- registration confirmation
- payment success
- payment failed
- reminder before event
- thank you after event

SMS / phone notification

Use SMS gateway integration for:

- payment confirmation
- registration success
- reminder one day before event
- urgent event updates

Example SMS:

"Your registration for [Event Name] is confirmed. Ticket No: XXXXX. Check your email for QR pass."

Ticket format

PDF or HTML-based ticket containing:

- event logo
- event name
- participant name
- designation

- category
- unique guest number
- QR code
- date/time/venue

8. Label / Sticker Printing Module

The same participant data and QR should be used for printing labels.

Label contents

- Name
- Designation
- Organization
- Category
- QR code

Label customization features

- width / height custom size
- portrait / landscape
- font size
- show/hide designation
- show/hide organization
- show/hide QR
- show/hide category color strip
- custom margins
- printer calibration
- multiple templates per event/category

Example use cases

- 2x4 inch sticker

- 3x4 inch badge label
- VIP badge design
- organizer label
- sponsor label

Admin should be able to print:

- one by one
- bulk print
- filtered print by category
- only unchecked labels
- reprint labels

Also log:

- printed/not printed
- printed at
- printed by

9. Dashboard & Admin Panel

Admin authentication

Username/password basis is okay, but it should be implemented securely.

Better recommendation

Use:

- email/username + password
- hashed passwords
- session management
- optional 2FA later

Admin roles

Recommended roles:

- Super Admin
- Event Admin
- Registration Staff
- Scanner Staff
- Finance/Admin Accounts
- Viewer/Read Only

This prevents everyone from accessing all features.

Dashboard widgets

Per event dashboard:

- total registrations
- total paid
- total unpaid
- total imported
- total check-ins
- lunch taken count
- dinner taken count
- card delivered count
- labels printed
- category-wise participant counts
- daily registration trend
- payment collection total
- pending payments

- event attendance percentage

Useful dashboard tables

- recent registrations
- recent scans
- payment logs
- failed transactions
- category summary
- imported participant list
- check-in status table

10. Reports & Exports

Admin should be able to download reports in:

- CSV
- Excel
- PDF summary

Export filters

- by event
- by category
- by payment status
- by check-in status
- by lunch/dinner status
- by date range
- by imported/self-registered source

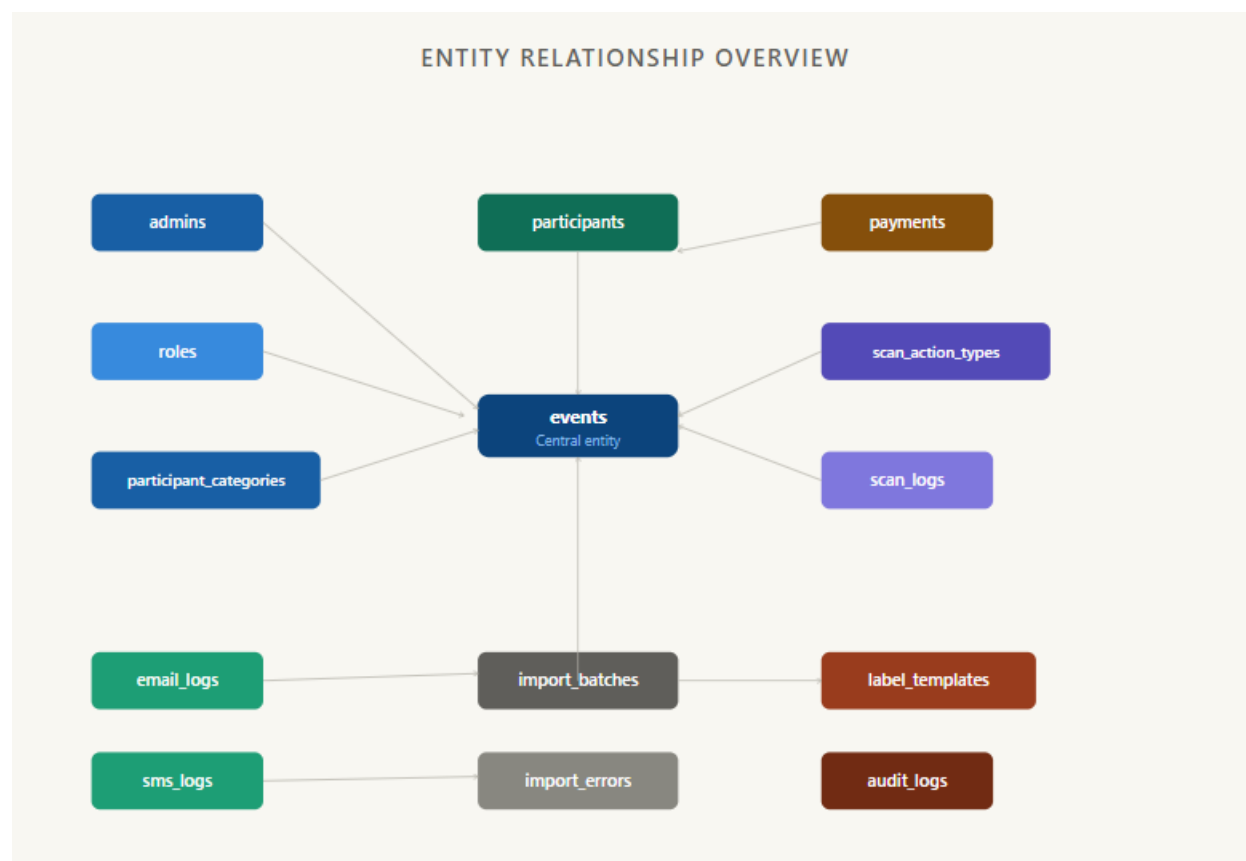
Important reports

1. full guest list

2. payment report
3. attendance report
4. meal consumption report
5. label printing report
6. card delivery report
7. category-wise summary
8. scanner activity report

11. Suggested Database Plan

A relational database is best. PostgreSQL is a strong choice.



Main Tables

1. admins

Stores admin users.

Fields:

- id
- username
- email
- password_hash
- full_name
- role_id
- is_active
- last_login
- created_at
- updated_at

2. roles

Stores admin roles.

Fields:

- id
- role_name
- permissions_json

3. events

Stores event details.

Fields:

- id
- name
- slug
- description
- venue

- start_datetime
- end_datetime
- registration_open_at
- registration_close_at
- banner_url
- logo_url
- status
- settings_json
- created_by
- created_at
- updated_at

4. participant_categories

Stores categories per event or reusable templates.

Fields:

- id
- event_id
- name
- description
- is_paid
- price
- currency
- badge_color
- sort_order
- is_active
- created_at

- updated_at

5. participants

Main participant table.

Fields:

- id
- event_id
- category_id
- full_name
- email
- phone
- designation
- organization
- address
- gender
- notes
- registration_source (self/csv/admin/manual)
- registration_status
- guest_number
- qr_token
- ticket_url
- label_printed
- created_at
- updated_at

6. payments

Stores payment records.

Fields:

- id
- participant_id
- event_id
- category_id
- amount
- currency
- payment_gateway
- gateway_transaction_id
- merchant_reference
- qr_payload
- payment_status
- paid_at
- raw_response_json
- created_at
- updated_at

7. scan_action_types

Customizable scan actions.

Fields:

- id
- event_id
- action_name
- action_code
- allow_multiple

- is_active
- sort_order
- created_at

Examples:

- CHECKIN
- LUNCH
- DINNER
- CARD_DELIVERY

8. scan_logs

Stores every QR scan action.

Fields:

- id
- event_id
- participant_id
- action_type_id
- scanned_by_admin_id
- scan_device
- scan_location
- remarks
- scanned_at
- created_at

9. email_logs

Tracks email sending.

Fields:

- id
- participant_id
- event_id
- email_type
- recipient_email
- subject
- status
- sent_at
- error_message

10. sms_logs

Tracks SMS sending.

Fields:

- id
- participant_id
- event_id
- sms_type
- recipient_phone
- message
- status
- sent_at
- error_message

11. import_batches

Tracks CSV imports.

Fields:

- id
- event_id
- imported_by
- file_name
- total_rows
- success_rows
- failed_rows
- created_at

12. import_errors

Stores failed CSV rows.

Fields:

- id
- batch_id
- row_number
- raw_data_json
- error_message

13. label_templates

Stores label settings.

Fields:

- id
- event_id
- template_name
- width
- height

- show_qr
- show_designation
- show_category
- config_json
- created_at

14. audit_logs

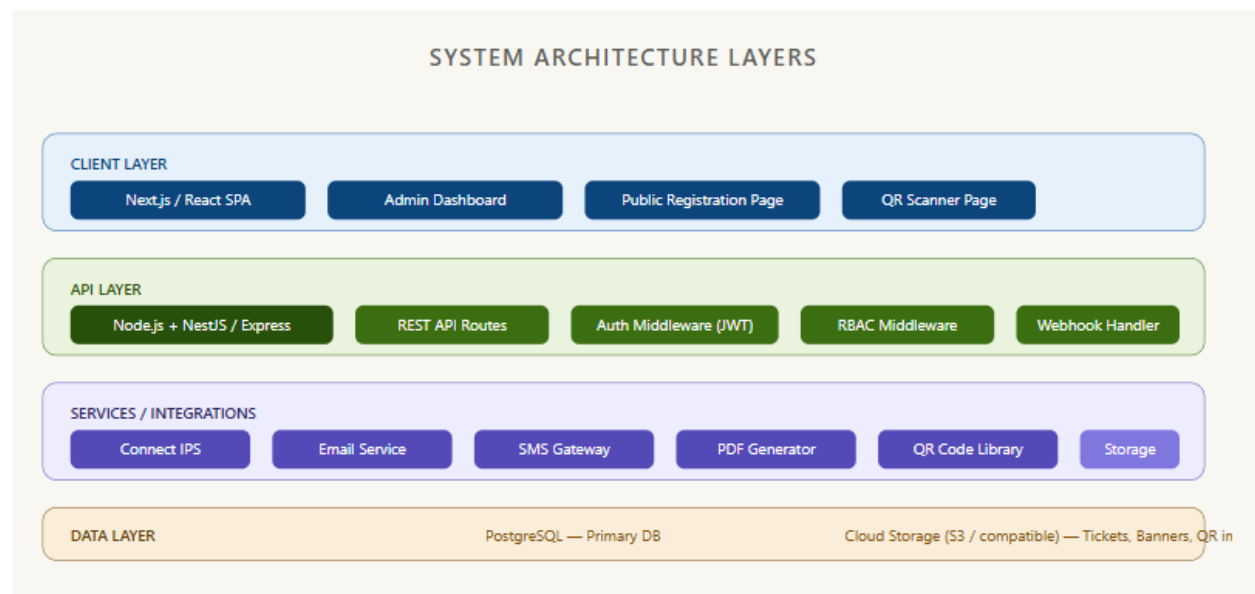
For admin actions.

Fields:

- id
- admin_id
- action
- entity_type
- entity_id
- old_value_json
- new_value_json
- created_at

This is important for accountability.

12. Recommended Architecture



Frontend

- Next.js or React-based web app
- Admin dashboard
- Public registration pages
- QR scan pages
- Label print pages

Backend

- Node.js with Express/NestJS or Next.js API routes
- REST API or modular backend services

Database

- PostgreSQL

File/Document storage

- Cloud storage for ticket PDFs, event banners, QR images, imports

Integrations

- Connect IPS payment API
- Email service

- SMS gateway
- PDF generation
- QR generation library

13. Admin Workflow Example

For a new event

1. Admin logs in
2. Creates event
3. Adds participant categories
4. Considers event is free or paid
5. If paid Sets ticket prices
6. Enables payment for selected categories
7. Creates scan action types
8. Selects label template
9. Publishes registration page
10. Imports sponsor/VIP/organizer CSV if needed
11. Monitors registrations/payments
12. Uses scanning during event day
13. Exports reports after event

14. Public User Workflow Example

1. User visits event page
2. Chooses category
3. Fills form
4. Pays through dynamic QR if event is paid.
5. Receives confirmation email/SMS

6. Receives QR ticket
7. Comes to event
8. QR scanned for check-in/lunch/dinner etc.

15. Security Plan

This system handles personal data and payments, so security is critical.

Must-have security controls

- password hashing
- HTTPS only
- CSRF protection
- rate limiting
- input validation
- file upload validation for CSV
- role-based access control
- signed secure QR token instead of plain predictable IDs
- payment callback signature verification
- audit logs
- backup strategy
- database encryption at rest if possible
- admin session timeout

Important logic protection

- QR must be event-specific
- participant from Event A must not be valid for Event B
- duplicate meal claims should be blocked
- payment must be verified server-side
- imported free users should not accidentally enter paid flow

Suggested Additional Improvements

These will make the platform much stronger.

- Capacity Management
- Promo Codes: Discount codes for selected categories.
- Approval-Based Registration: Some categories may require admin approval before ticket issuance.
- Companion / Group Booking: One user can register multiple guests.
- Onsite Manual Registration Desk: Admin/staff can register walk-in participants on event day.
- Badge Status Track: *not printed, printed, collected*
- Attendance Analytics: Heatmap of arrival time, category-wise attendance rate.
- Certificate Eligibility: If event later needs certificates, check-in logs can be reused.
- Multi-language Support: English/Nepali support in forms and badges.
- WhatsApp/Viber optional notifications
- Offline-friendly scan mode